COMPETENCY STANDARD

FOR

SHIP PAINTING

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.
Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from specific industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (50 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SHP-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS Policies and Procedures 2. Apply Personal Health and Safety Practices 3. Report Hazards and Risks 4. Respond to Emergencies.	20
SICIP-SBD-SHP-02-G	Carry Out Workplace Interaction	1 Interpret Workplace Communication and Etiquette 2 Read and Understand Workplace Documents 3 Participate in Workplace Meetings and Discussions 4 Practice Professional Ethics at Work	20
SICIP-SBD-SHP-03-G	Operate in a Team Environment	1 Identify Team Goals and Work Processes 2 Identify Own Role and Responsibilities Within Team 3 Communicate and Co-Operate with Team Members 4 Practice Problem Solving Within the Team	10
		Total Hour	**50 Hrs.**

Sector Specific Competencies (Total 40 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SHP-01-S	Apply Green Practices in Ship Building Sector	1. Interpret Green Concepts 2. Minimize Resource use in the Workplace 3. Implement Waste Management Practices	15
SICIP-SBD-SHP-02-S	Work with Hand Tools and Power Tools	1. Identify and Inspect Hand and Power Tools 2. Use Hand Tools Properly and Safely 3. Operate Power Tools Properly and Safely 4. Clean and Maintain Hand and Power Tools	25
		Total Hour	**40 Hrs.**

Occupation Specific Competencies (Total 270 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-SHP-01-O	Identify Basic Ship Painting Works	1. Identify Major Structural Parts of a Ship 2. Identify Basic Ship Painting Requirements. 3. Identify Surface Areas for Cleaning and Painting. 4. Identify Types of Paint and Painting Processes.	30
SICIP-SBD-SHP-02-O	Carry Out Surface Preparation	1. Prepare for Work. 2. Remove Slag and Contamination 3. Perform Surface Cleaning. 4. Clean and Maintain Workplace.	60
SICIP-SBD-SHP-03-O	Apply Primer Coat to Hull	1. Prepare for Work. 2. Perform Primer Coat Application. 3. Clean and Maintain Workplace.	50
SICIP-SBD-SHP-04-O	Perform Tie and Anti-Fouling Coat to Underwater Hull	1. Prepare for Work. 2. Perform Tie-Coat Works. 3. Apply Anti-Fouling Coat. 4. Perform Rectification Work. 5. Clean and Maintain Workplace.	60
SICIP-SBD-SHP-05-O	Perform Top Coat to Above Water Hull, Superstructure and Other Areas	1. Prepare for Work. 2. Apply Top Coat. 3. Carry Out Rectification Work. 4. Clean and Maintain Workplace.	70
		Total Hours	**270 Hrs.**

COMPETENCY STANDARD: SHIP PAINTING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SHP-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **OHS policies** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including **personal protective equipment (PPE).** 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 **Emergency response plans and procedures** are responded to. 4.3 **First aid procedures** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace documents, signs and symbols 4.3 Codes of conduct 4.4 Projector

	4.5 Learning manual 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT WORKPLACE INTERACTION**	**Nominal Duration:** 20 hrs.	**Unit Code:** SICIP-SBD-SHP-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **courteous manner** to gather and convey information. 1.4 **Workplace procedures and matters** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **appropriate sources**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Demonstrating active participation in workplace meeting 2.4 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-SBD-SHP-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through **sharing information** or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Sharing information	1.1. Agenda 1.2. Minutes

	1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR**	**Nominal Duration:** 15 hrs.	**Unit Code:** SICIP-SBD-SHP-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **Green concepts in the shipbuilding sector** are understood. 1.2 **Energy-efficient technologies** are recognized. 1.3 **Use of eco-friendly materials** is understood. 1.4 **Pollution control measures** are interpreted. 1.5 **Regulatory compliance** with regulations and conventions is identified. 1.6 **Energy recovery systems** are identified as methods to improve energy efficiency.
2. Minimize resources use in the workplace	2.1 **Resource usage in shipbuilding processes** is identified. 2.2 Water consumption methods are identified through the installation of efficient water management systems. 2.3 **Process improvements** are understood. 2.4 Compliance with environmental regulations and industry standards is interpreted.
3. Implement waste management practices	3.1 **The types of waste in shipbuilding** are identified and described. 3.2 **Environmental impacts of improper waste disposal** in shipbuilding are understood. 3.3 **Principles of waste hierarchy** are understood for sustainable waste management. 3.4 **Importance of safe storage and containment of waste materials** is understood.

Range of Variables

Variable	Range May include but not limited to:
1. Green concepts in the shipbuilding sector	1.1 Eco-friendly Hull Design 1.2 Use of Sustainable and Recyclable Materials 1.3 Waste Reduction and Management Practices

	1.4 Emission Control and Air Pollution Reduction 1.5 Ballast Water Treatment and Marine Ecosystem Protection
2. Energy-efficient technologies	2.1 Hybrid and Electric Propulsion Systems 2.2 Advanced Hull Coatings to Reduce Drag 2.3 Waste Heat Recovery Systems 2.4 Air Lubrication Systems 2.5 Energy-efficient LED Lighting and Automation 2.6 Smart Energy Management and Monitoring Systems
3. Use of eco-friendly materials	3.1 Recycled Steel and Aluminum 3.2 Biodegradable Composites and Polymers 3.3 Sustainably Sourced Timber 3.4 Non-toxic Anti-fouling Coatings 3.5 Natural Fiber Reinforced Plastics 3.6 Low VOC (Volatile Organic Compounds) Paints and Sealants
4. Pollution control measures	4.1 Prevention of water 4.2 Prevention of air 4.3 Prevention of noise pollution
5. Regulatory compliance	5.1 International Maritime Organization (IMO) Regulations 5.2 Environmental Protection Agency (EPA) Standards 5.3 Marine Pollution (MARPOL) Convention Compliance 5.4 Occupational Health and Safety (OHS) Regulations 5.5 Ship Recycling and Basel Convention Guidelines 5.6 Local and National Environmental Permitting
6. Energy recovery systems	6.1 Waste Heat Recovery Boilers (WHRB) 6.2 Thermoelectric Energy Conversion Systems 6.3 Exhaust Gas Recirculation (EGR) Systems 6.4 Regenerative Braking for Hybrid Propulsion
7. Resource usage in shipbuilding processes	7.1 Raw Material Consumption (Steel, Aluminum, Composites) 7.2 Water Usage in Fabrication and Painting 7.3 Energy Consumption During Manufacturing 7.4 Use of Recycled and Sustainable Materials
8. Process improvements	8.1 Automation and Robotics in Assembly 8.2 Modular Construction and Prefabrication 8.3 Digital Twin and 3D Modeling Integration 8.4 Just-In-Time (JIT) Inventory Management 8.5 Enhanced Quality Control and Testing Procedures
9. Types of waste in shipbuilding	9.1 Hazardous wastes 9.1.1 Paints 9.1.2 Solvents 9.1.3 Oils 9.1.4 Asbestos 9.2 Non-hazardous wastes 9.2.1 Metal scraps 9.2.2 Wood and general refuse

10. Environmental impacts of improper waste disposal	10.1 Including soil contamination 10.2 Water pollution, and air quality degradation, are understood.
11. Principles of waste hierarchy	11.1 Reduction 11.2 Reuse 11.3 Recycling 11.4 Recovery 11.5 Disposal
12. Importance of safe storage and containment of waste materials	12.1 Prevent leaks 12.2 Prevent spills 12.3 Environmental contamination

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental Green Concepts and Sustainability Principles in Shipbuilding 1.2 Overview of Energy-Efficient Technologies and Their Applications 1.3 Types and Benefits of Eco-Friendly Materials in Shipbuilding 1.4 Key Pollution Control Measures and Their Implementation 1.5 Regulatory Frameworks, Conventions, and Compliance Requirements 1.6 Energy Recovery Systems and Their Role in Enhancing Efficiency 1.7 Resource and Water Usage Management in Shipbuilding Processes 1.8 Understand of environmental Impacts, and Sustainable Practices
2. Underpinning Skills	2.1 Applying Green Concepts in Shipbuilding Practices 2.2 Evaluating and Selecting Energy-Efficient Technologies 2.3 Identifying and Using Eco-Friendly Materials Effectively 2.4 Implementing Pollution Control Measures Onsite 2.5 Interpreting and Ensuring Regulatory Compliance 2.6 Operating and Maintaining Energy Recovery Systems 2.7 Monitoring and Managing Resource and Water Usage 2.8 Improving Shipbuilding Processes for Environmental Sustainability 2.9 Classifying, Handling, and Safely Storing Shipbuilding Waste
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **USE HAND AND POWER TOOLS**	**Nominal Duration:** 25 hrs.	**Unit Code:** SICIP-SBD-SHP-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate **<u>hand tools</u>** are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 **<u>Measuring tools</u>** are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to

	storage. 4.5. Defective hand and power tools are inspected and repaired or replaced. 4.6. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Measuring tape 3.3 Hose level 3.4 Water level 3.5 Calipers 3.6 Steel rule 3.7 Meter rule 3.8 Spirit level 3.9 Protractor 3.10 Tri-square 3.11 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification

	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY BASIC SHIP PAINTING WORKS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-SBD-SHP-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify basic ship painting works. It specifically includes the task of identifying major structural parts of a ship, identifying basic painting requirements, identifying surface areas for cleaning and painting and identifying types of paint and painting processes.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify major structural parts of a ship	1.1. **<u>Major structural parts of the ship</u>** are identified from ship layout diagrams or physical observation. 1.2. Functions of key structural components are described. 1.3. Access routes are identified for their role in emergency evacuation and fire response.
2. Identify basic painting requirements	2.1 Principles of ship painting are identified and described. 2.2 Class rules and environmental requirements are identified. 2.3 Common ship painting **<u>terminology</u>** is identified and interpreted.
3. Identify surface areas for cleaning and painting	3.1 **<u>Ship surface</u>** for painting is identified. 3.2 **<u>Methods for preparing surface</u>** are identified. 3.3 **<u>Confined spaces</u>** requiring painting are identified.
4. Identify types of paint and painting processes	4.1 **<u>Types of paint</u>** are identified. 4.2 Painting procedure and precautions are identified. 4.3 Painting **<u>application methods</u>** are identified.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Major structural parts of the ship	1.1 Hull 1.2 Keel 1.3 Deck 1.4 Accommodation 1.5 Hatch and hatch cover 1.6 Tanks 1.7 Anchor chain and winch 1.8 Anchor 1.9 Superstructure 1.10 Deck crane 1.11 Mast 1.12 Bridge

	1.13 Funnel 1.14 Rudder 1.15 Bollard
2. Terminology	2.1 Antifouling 2.2 Primer 2.3 Topcoat 2.4 Striping 2.5 Fendering 2.6 Rust Converter 2.7 Varnish 2.8 Galvanizing 2.9 Epoxy 2.10 Blasting 2.11 Chalking 2.12 Overcoating 2.13 Undercoating 2.14 Dry Film Thickness (DFT) 2.15 Wet Film Thickness (WFT) 2.16 Volume solid (VS) 2.17 Loss factor
3. Ship surface	3.1 External parts: 3.1.1 Underwater 3.1.2 Boot top zone 3.1.3 Topside (above boot top) 3.1.4 Deck and super structure 3.2 Internal /confined parts: 3.2.1 Cargo hold 3.2.2 Engine room 3.2.3 Oil tank 3.2.4 Accommodation 3.2.5 DB tank 3.2.6 Ballast tank 3.2.7 Fresh water tank 3.2.8 Sea water tank 3.2.9 Wing tank 3.2.10 Deep tank 3.2.11 Chain locker 3.2.12 Fore and after peak
4. Methods for preparing surfaces	4.1 Abrasive Blasting 4.2 Wire Brushing 4.3 Manual Cleaning 4.4 Chemical Cleaning 4.5 Water Blasting 4.6 Disc sanding and cup brushing 4.7 Sandpapering 4.8 Sweeping
5. Confined spaces	5.1 Tanks

	5.2 Cargo Holds 5.3 Ballast Tanks 5.4 Cargo Pits 5.5 Double Bottoms 5.6 Pipe Voids 5.7 Ventilation Ducts 5.8 Bilge Areas 5.9 Mooring Pits 5.10 Engine Rooms 5.11 Fuel Tanks 5.12 Tunnels 5.13 Pump Rooms 5.14 Storage Lockers
6. Types of paint	6.1 Antifouling Paint 6.2 Epoxy Paint 6.3 Polyurethane Paint 6.4 Alkyd Paint 6.5 Zinc-Rich Primer 6.6 Acrylic Paint 6.7 Vinyl Paint 6.8 Coal Tar Epoxy 6.9 Polyurea Coating 6.10 Silicone-Based Paint
7. Application methods	7.1 Brushing 7.2 Rolling 7.3 Air spraying

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify major structural parts of the ship from layout diagrams or physical observation. 1.2 Describe the functions of key structural components. 1.3 Identify access routes for emergency evacuation and fire response. 1.4 Understand the SOLAS Convention as the international standard for ship safety. 1.5 Explain the main objectives of SOLAS. 1.6 Interpret SOLAS rules on fire and safety. 1.7 Identify fire detection and alarm systems as per FSS Code requirements. 1.8 Identify fire hydrants, hoses, and nozzles in the ship's fire main system. 1.9 Recognize emergency escape breathing devices (EEBD) and firefighter's outfits. 1.10 Describe principles of ship painting and environmental requirements. 1.11 Understand types of paint, painting procedures, and precautions.

	1.12 Describe painting application methods for effective coverage.
2. Underpinning Skills	2.1 Performing identification of major structural parts of a ship from layout diagrams or physical observation. 2.2 Demonstrating description of functions of key structural components. 2.3 Applying knowledge to identify access routes for emergency evacuation and fire response. 2.4 Performing identification of ship painting principles and requirements. 2.5 Demonstrating understanding of class rules and environmental requirements for painting. 2.6 Applying common ship painting terminology and interpreting its meaning. 2.7 Identifying surface areas for cleaning and painting, including confined spaces. 2.8 Demonstrating preparation methods for surfaces before painting. 2.9 Performing identification of types of paint, painting procedures, precautions, and application methods.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 workplace (actual or simulate) 4.2 materials relevant to proposed activities 4.3 tools, equipment and documentation 4.4 specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified major structural parts of a ship 1.2 identified basic ship painting requirements. 1.3 identified surface areas for cleaning and painting. 1.4 identified types of paint and painting processes.
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT SURFACE CLEANING**	Nominal Duration: 60 Hrs.	Unit Code: SICIP-SBD-SHP-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out surface cleaning. It specifically includes the task of preparing for work, performing surface cleaning, and cleaning and maintaining workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for work	1.1 Job specifications and instructions are read and interpreted. 1.2 Appropriate **personal protective equipment (PPE)** is identified and selected. 1.3 Appropriate **tools and equipment** are identified and selected. 1.4 Appropriate **materials** are identified and selected. 1.5 Selected tools, equipment and materials are prepared as per job requirement.
2. Perform surface cleaning	2.1 **Surface cleaning requirements** are identified and followed to ensure the proper adhesion of new paint. 2.2 **Sludge and contaminants** are removed from the surface to prepare it for painting. 2.3 Surface area is cleaned as per job requirements. 2.4 Cleaned surfaces are inspected to confirm they are free from debris and contamination. 2.5 Cleaning results are verified to ensure compliance with quality and safety standards.
3. Clean and maintain workplace	3.1 Tools and equipment are cleaned and maintained as per standard operating procedure. 3.2 Tools and equipment are safely and securely stored. 3.3 Workplace is cleaned and waste material disposed of.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal protective equipment (PPE)	1.1. Safety harness 1.2. Safety googles 1.3. Ear plugs 1.4. Hand gloves 1.5. Apron (with respiratory air-fed blast hood) 1.6. Chemical resistant gas mask 1.7. Safety shoes 1.8. External air feed 1.9. Gas mask with torch light

2. Tools and equipment	2.1 Hand tools: 2.1.1 Chipping hammer 2.1.2 Chisel 2.1.3 Wire brush 2.1.4 Scrappers and blades 2.1.5 Scrubbing brushes 2.1.6 Extension poles 2.2 Power tools: 2.2.1 Grinder 2.2.2 Needle gun 2.2.3 Sand blaster 2.2.4 Bristle blaster 2.2.5 Air blower 2.3 Measuring tools: 2.3.1 Coat thickness gauge 2.3.2 Wet film thickness gauge 2.3.3 Surface profile gauge 2.3.4 Holiday detector 2.3.5 Salt contamination meter 2.3.6 Pull-off test machine 2.3.7 Surface temperature gauge 2.3.8 Hygrometer 2.4 Equipment: 2.4.1 Shot blasting machine 2.4.2 Airless spray-painting machine 2.4.3 Oil-free air compressor 2.4.4 Vacuum cleaner 2.4.5 High pressure water jet 2.4.6 High pressure washing machine
3. Materials	3.1 Cleaning solvents 3.2 Emery paper 3.3 Cotton wastage
4. Surface cleaning requirements	4.1 Hand Tool Cleaning 4.2 Power Tool Cleaning 4.3 Near-White Blast Cleaning 4.4 High Pressure Water Cleaning 4.5 Solvent Cleaning 4.6 Brush-Off Blast Cleaning
5. Sludge and contaminants	5.1 **Sludge:** 5.1.1 Paint Sludge 5.1.2 Rust Sludge 5.1.3 Fuel Oil Sludge 5.1.4 Grease Sludge 5.1.5 Wastewater Sludge 5.2 **Contaminants:** 5.2.1 Rust 5.2.2 Oil & Grease

	5.2.3 Salt 5.2.4 Dirt & Dust 5.2.5 Mill Scale 5.2.6 Old Paint 5.2.7 Corrosion Products

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate personal protective equipment (PPE). 1.2 Identify appropriate tools and equipment. 1.3 Identify surface cleaning requirements to ensure proper paint adhesion. 1.4 Remove sludge and contaminants from surfaces to prepare for painting. 1.5 Clean surface areas according to job specifications. 1.6 Inspect cleaned surfaces to confirm they are free from debris and contamination. 1.7 Verify cleaning results to ensure compliance with quality and safety standards. 1.8 Clean and maintain tools and equipment according to standard operating procedures. 1.9 Store tools and equipment safely and securely. 1.10 Clean the workplace and dispose of waste materials appropriately.
2. Underpinning Skills	2.1 Performing reading and interpretation of job specifications and instructions. 2.2 Demonstrating identification and selection of appropriate personal protective equipment (PPE). 2.3 Maintaining preparation of selected tools, equipment, and materials as per job requirements. 2.4 Performing surface cleaning to remove sludge and contaminants for proper paint adhesion. 2.5 Inspecting cleaned surfaces to confirm they are free from debris and contamination. 2.6 Verifying cleaning results to ensure compliance with quality and safety standards. 2.7 Cleaning and maintaining tools and equipment according to standard operating procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulate) 4.2 Materials relevant to proposed activities 4.3 Tools, equipment and documentation 4.4 Specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for work. 1.2 removed slag and contamination 1.3 performed surface cleaning. 1.4 cleaned and maintained workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY PRIMER COAT TO STRUCTURE**	**Nominal Duration:** 50 Hrs**.**	**Unit Code:** SICIP-SBD-SHP-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply primer coat to structure. It specifically includes the task of preparing for work, performing primer coat application, and cleaning and maintaining workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Prepare for work	1.1. Job specifications and instructions are read and interpreted. 1.2. Appropriate personal protective equipment (PPE) is identified and selected. 1.3. Appropriate **tools and equipment** are identified and selected. 1.4. Appropriate **materials** are identified and selected. 1.5. Selected tools, equipment and materials are prepared as per job requirement.
2. Perform primer coat application	2.1 Work area is set-up as per job requirement. 2.2 **Types of coating** are identified and selected as per technical data sheet. 2.3 Surface is cleaned and dried to remove grease, oil and salt contamination. 2.4 Adequate ventilation is established as per standard operating procedure. 2.5 Humidity, temperature and dew point are identified and ensured as per job requirement. 2.6 Primer coat on internal and external areas is applied as per job specification. 2.7 Overcoating and wet film thickness (WFT) are checked as per technical data sheet.
3. Clean and maintain workplace	3.1. Tools and equipment are cleaned and maintained as per standard operating procedure. 3.2. Tools and equipment are safely and securely stored. 3.3. Workplace is cleaned and waste material disposed of.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Tools and equipment	1.1. Tools: 1.1.1. Ladder 1.1.2. Platform 1.1.3. Paint brushes 1.1.4. Spatula

	1.1.5. Rollers 1.1.6. Extension poles 1.2. Equipment: 1.2.1. Airless sprayer/gun 1.2.2. Pumps 1.2.3. Stirrers 1.2.4. Scaffold 1.2.5. Ladder
2. Materials	2.1 Base paint 2.2 Curing agent 2.3 Solvent 2.4 Thinner 2.5 Waste cotton
3. Types of coating	3.1 Epoxy Coatings 3.2 Polyurethane Coatings 3.3 Acrylic Coatings 3.4 Zinc-rich Coatings 3.5 Alkyd Coatings 3.6 Silicone Coatings 3.7 Anti-fouling Coatings 3.8 Anti-corrosive Coatings

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the appropriate tools and equipment. 1.2 Identify appropriate materials. 1.3 Set up the work area according to job requirements. 1.4 Identify types of coating based on the technical data sheet. 1.5 Establish adequate ventilation according to standard operating procedures. 1.6 Identify humidity, temperature, and dew point are suitable for painting. 1.7 Primer coat on internal and external areas as per job specifications. 1.8 Overcoating and wet film thickness (WFT) according to the technical data sheet.
2. Underpinning Skills	2.1 Maintaining preparation of selected tools, equipment, and materials as per job requirements. 2.2 Performing setup of the work area according to job requirements. 2.3 Applying primer coat on internal and external surfaces. 2.4 Demonstrating checking of overcoating and wet film thickness (WFT) according to technical data sheet. 2.5 Cleaning and maintaining tools and equipment as per standard operating procedures. 2.6 Maintaining workplace cleanliness and disposing of waste materials safely.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulate) 4.2 Materials relevant to proposed activities 4.3 Tools, equipment and documentation 4.4 Specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for work. 1.2 performed primer coat application. 1.3 cleaned and maintained workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM TIE AND ANTI-FOULING COAT TO UNDERWATER HULL**	**Nominal Duration:** 60 Hrs**.**	**Unit Code:** SICIP-SBD-SHP-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform tie and anti-fouling coat to undercoat hull. It specifically includes the task of preparing for work, performing tie-coat works, applying anti-fouling coat, carrying out rectification work and cleaning and maintaining workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for work	1.1. Job specifications and instructions are read and interpreted. 1.2. Appropriate personal protective equipment (PPE) is identified and selected. 1.3. Appropriate **tools and equipment** are identified and selected. 1.4. Appropriate **materials** are identified and selected. 1.5. Selected tools, equipment and materials are prepared as per job requirement.
2. Perform tie-coat works	2.1 Underwater hull surface is cleaned and prepared properly to ensure the tie coating. 2.2 Tie-coat works are carried out to bond the primer with anti-fouling paint as per specified procedures. 2.3 Tie coat is inspected and cleaned before the application of anti-fouling paint. 2.4 Quality of tie-coat work is checked to confirm proper bonding and surface preparation.
3. Apply anti-fouling coat	3.1 **Application conditions** are checked and maintained as per standard operating procedures. 3.2 Surface preparation is carried out before applying the anti-fouling coat. 3.3 Anti-fouling coat is applied in accordance with the guidelines and specifications. 3.4 Standard curing time and temperature are monitored and maintained. 3.5 Work area is maintained clean and free from contaminants.
4. Carry out rectification work	4.1 **Paint defects** are identified and assessed to determine the appropriate **rectification method**. 4.2 Paint rectification is performed following standard operating procedures. 4.3 Rectified anti-fouling surface is inspected and checked

	with quality assurance standards and specifications. 4.4 Final finishing coat is applied as per job requirements. 4.5 Final finish is inspected to confirm that the rectified surface meets the necessary performance.
5. Clean and maintain workplace	5.1. Tools and equipment are cleaned and maintained as per standard operating procedure. 5.2. Tools and equipment are safely and securely stored. 5.3. Workplace is cleaned and waste material disposed of.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Tools and equipment	1.1. Tools: 1.1.1. Spatula 1.1.2. Extension poles 1.1.3. Paint brushes 1.1.4. Roller brushes 1.2. Equipment: 1.2.1. Airless sprayer 1.2.2. Stirrers 1.2.3. Air dusters 1.2.4. Scaffold
2. Materials	2.1 Anti-fouling paint 2.2 Thinner 2.3 Masking tape 2.4 Waste cotton
3. Application conditions	3.1 Temperature 3.1.1 Surface 3.1.2 Ambient 3.2 Relative Humidity 3.3 Weather Conditions 3.4 Surface Moisture 3.5 Drying Time 3.6 Application Method 3.7 Ventilation 3.8 Coating Thickness 3.9 Product Compatibility
4. Paint defects	4.1 Blistering 4.2 Peeling/Flaking 4.3 Cracking/Crazing 4.4 Chalking 4.5 Orange Peel 4.6 Runs and Sags 4.7 Fisheyes 4.8 Mottling 4.9 Discoloration 4.10 Rust Bleeding

	4.11 Poor Adhesion 4.12 Solvent Pop 4.13 Streaking 4.14 Wrinkling 4.15 Under-Curing
5. Rectification method	5.1 Sanding and Cleaning 5.2 Scraping and Reapplying Primer/Topcoat 5.3 Flexible Paint Application 5.4 UV-Resistant Reapplication 5.5 Sanding and Correct Spraying Technique 5.6 Thinner Coat Reapplication 5.7 Contamination Cleaning and Compatibility Check 5.8 Uniform Coverage Reapplication 5.9 UV-Resistant Reapplication 5.10 Rust Removal and Anti-Corrosive Treatment 5.11 Primer and Topcoat Application 5.12 Ventilation and Proper Thinning 5.13 Even Application Technique 5.14 Proper Drying Conditions and Reapplication 5.15 Curing and Reapplication

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment. 1.2 Identify and select appropriate materials. 1.3 Comprehend the underwater hull surface for tie-coat application. 1.4 Recognize tie-coat works to bond the primer with anti-fouling paint following specified procedures. 1.5 Describe the quality of tie-coat work to ensure proper bonding and surface preparation. 1.6 Describe application conditions for the anti-fouling coat according to standard operating procedures. 1.7 Explain the surface before applying the anti-fouling coat. 1.8 Identify anti-fouling coat in accordance with guidelines and specifications. 1.9 Identify curing time and temperature during the anti-fouling process. 1.10 Identify paint defects and assess for appropriate rectification methods.
2. Underpinning Skills	2.1 Applying knowledge to identify and prepare tools, equipment, and materials as per job requirements. 2.2 Performing cleaning and preparation of underwater hull surfaces for tie-coat application. 2.3 Applying tie-coat to bond primer with anti-fouling paint and inspecting for proper surface preparation. 2.4 Performing application of anti-fouling coat, ensuring

	correct surface preparation, curing time, and environmental conditions. 2.5 Demonstrating rectification of paint defects following standard procedures. 2.6 Applying the final finishing coat. 2.7 Maintaining quality checks.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulate) 4.2 Materials relevant to proposed activities 4.3 Tools, equipment and documentation 4.4 Specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for work. 1.2 performed tie-coat works. 1.3 applied anti-fouling coat. 1.4 performed rectification work. 1.5 cleaned and maintained workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM TOP COAT TO ABOVE WATER HULL, SUPERSTRUCTURE AND OTHER AREAS**	Nominal Duration: 70 Hrs.	Unit Code: SICIP-SBD-SHP-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform top coat to above water hull, superstructure and other areas. It specifically includes the task of preparing for work, applying top coat, carrying out rectification work, and cleaning and maintaining workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for work	1.1. Job specifications and instructions are read and interpreted. 1.2. Appropriate personal protective equipment (PPE) is identified and selected. 1.3. Appropriate **tools and equipment** are identified and selected. 1.4. Appropriate **materials** are identified and selected. 1.5. Selected tools, equipment and materials are prepared as per job requirement.
2. Apply top coat	2.1 Surface is cleaned and prepared for top coat application to above water hull, superstructure, and **other areas**. 2.2 **Precautionary measures** are undertaken for work in confined spaces as per standard operating procedure. 2.3 Top coat works are carried out for bonding with primer coat as per technical data sheet. 2.4 Overcoating time, humidity and temperature are monitored and maintained as per standard operating procedure. 2.5 Standard curing time and temperature are monitored and maintained to ensure required paint hardness and bonding.
3. Carry out rectification work	3.1 Paint defects are identified and assessed to determine the appropriate rectification method. 3.2 Paint rectification is performed following standard operating procedures. 3.3 Rectified surface is inspected and checked with quality assurance standards and specifications. 3.4 Final finishing coat is applied as per job requirements. 3.5 Final finish is inspected to confirm that the rectified surface meets the necessary performance.
4. Clean and maintain workplace	4.1 Tools and equipment are cleaned and maintained as per standard operating procedure. 4.2 Tools and equipment are safely and securely stored.

	4.3 Workplace is cleaned and waste material disposed of.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Tools and equipment	1.1 Tools: 1.1.1 Spatula 1.1.2 Extension poles 1.1.3 Paint brushes 1.1.4 Roller brushes 1.2 Equipment: 1.2.1 Airless sprayer 1.2.2 Stirrers 1.2.3 Air dusters 1.2.4 Scaffold
2. Materials	2.1 Anti-fouling paint 2.2 Thinner 2.3 Masking tape 2.4 Waste cotton
3. Other areas	3.1 Deck 3.2 Mast 3.3 Derrick 3.4 Winch 3.5 Chain locker 3.6 DB tank 3.7 Cargo hold 3.8 Engine room 3.9 Oil tank 3.10 Fore and after peak tank
4. Precautionary measures	4.1 Free from gas 4.2 Free for oil, grease and fat 4.3 Free from chemicals 4.4 Clean and dry

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify appropriate tools and equipment. 1.2 Identify appropriate materials. 1.3 Identify surfaces for top coat application on above water hull, superstructure, and other areas. 1.4 Describe precautionary measures for work in confined spaces according to standard operating procedures. 1.5 Explain monitor overcoating time, humidity, and temperature to maintain proper conditions. 1.6 Describe standard curing time and temperature for required paint hardness and bonding. 1.7 Identify paint defects and assess for appropriate rectification methods.

	1.8 Identify paint rectification following standard operating procedures. 1.9 Explain the surface to verify compliance with quality standards.
2 Underpinning Skills	2.1 Applying knowledge to identify and prepare tools, equipment, and materials as per job requirements. 2.2 Performing cleaning and preparation of underwater hull surfaces for tie-coat application. 2.3 Applying tie-coat to bond primer with anti-fouling paint and inspecting for proper surface preparation. 2.4 Performing application of anti-fouling coat. 2.5 Demonstrating rectification of paint defects following standard procedures. 2.6 Applying the final finishing coat. 2.7 Maintaining quality checks to ensure proper bonding, surface preparation, and performance standards.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulate) 4.2 Materials relevant to proposed activities 4.3 Tools, equipment and documentation 4.4 Specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for work. 1.2 applied top coat. 1.3 carried out rectification work. 1.4 cleaned and maintained workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Ship Painting
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/laptop	01
2.	Multimedia projector	01
3.	Internet connectivity	01
4.	Spray Guns (Airless & Conventional)	05
5.	Paint Brushes	25
6.	Paint Rollers	25
7.	Paint Trays	25
8.	Air Compressors	02
9.	Grinders	10
10.	Scaffolding and Ladders	02
11.	Pressure Washers (Water)	02
12.	Paint Mixers	01
13.	Paint Pumps	02
14.	Paint Thickness Gauge	02
15.	Air blower	02
16.	Surface Cleaning Equipment (Blasting, Sandblasting Machines)	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Ship Painting the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.